

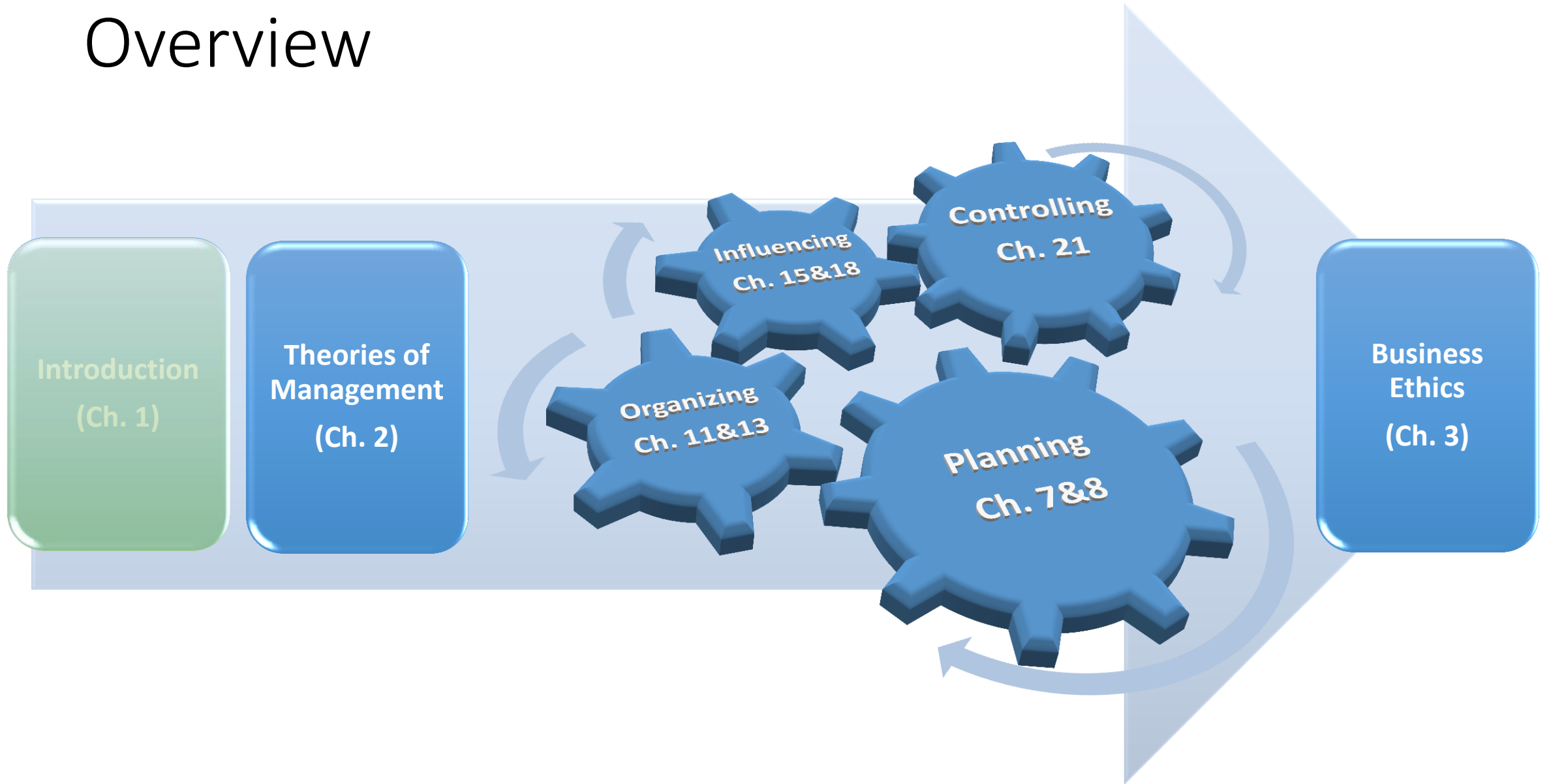
Chapter 2

History and Current Thinking

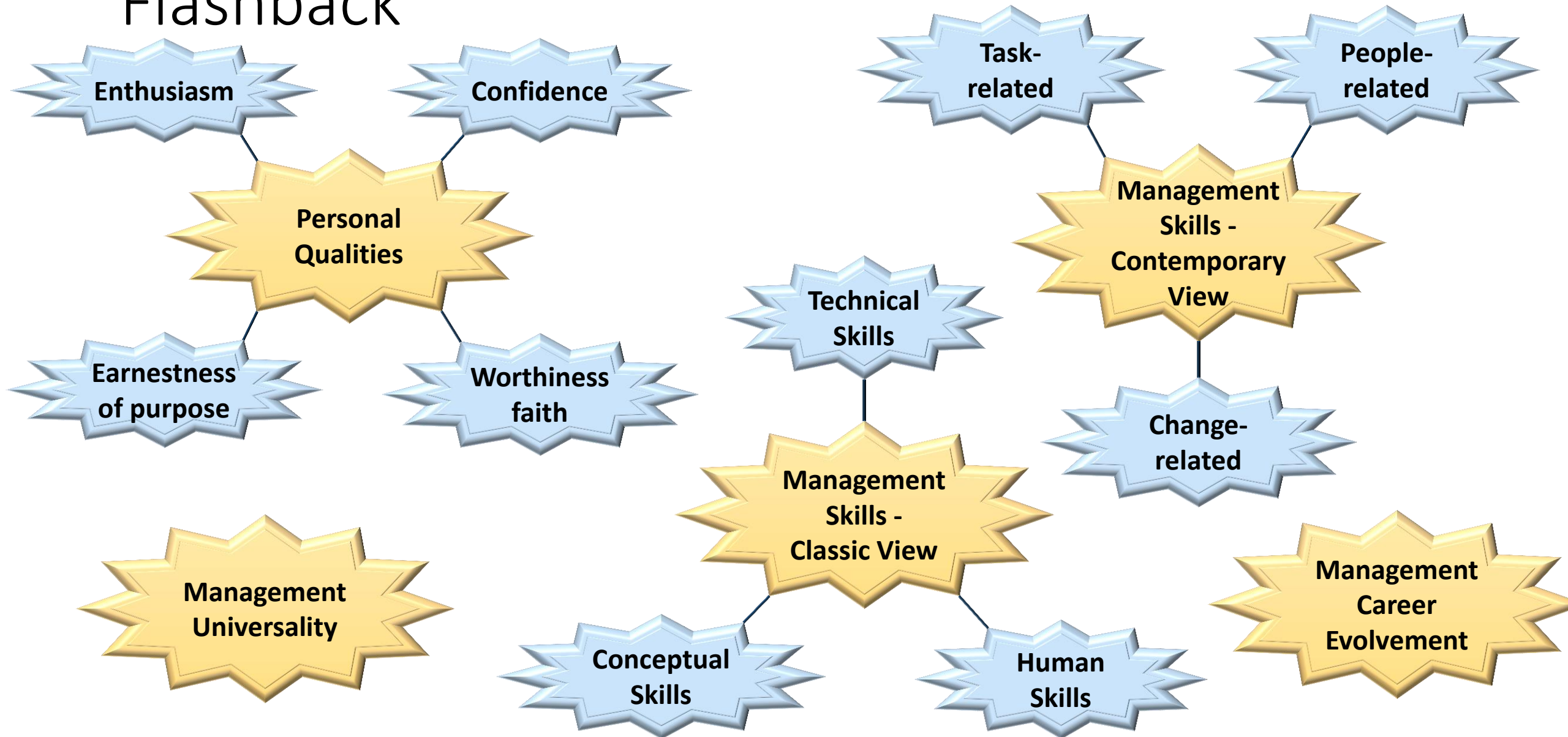
By

Dr. Doaa Saleh

Overview



Flashback



Objectives

1. *An understanding of the classical approach to management*
2. *An appreciation for the work of Frederick W.Taylor, Frank and Lillian Gilbreth, Henry L. Gantt, and Henri Fayol*
3. *An understanding of the behavioral approach to management*
4. *An understanding of the studies at the Hawthorne Works and the human relations movement*
5. *An understanding of the management science approach and how it has evolved*

Management Approaches

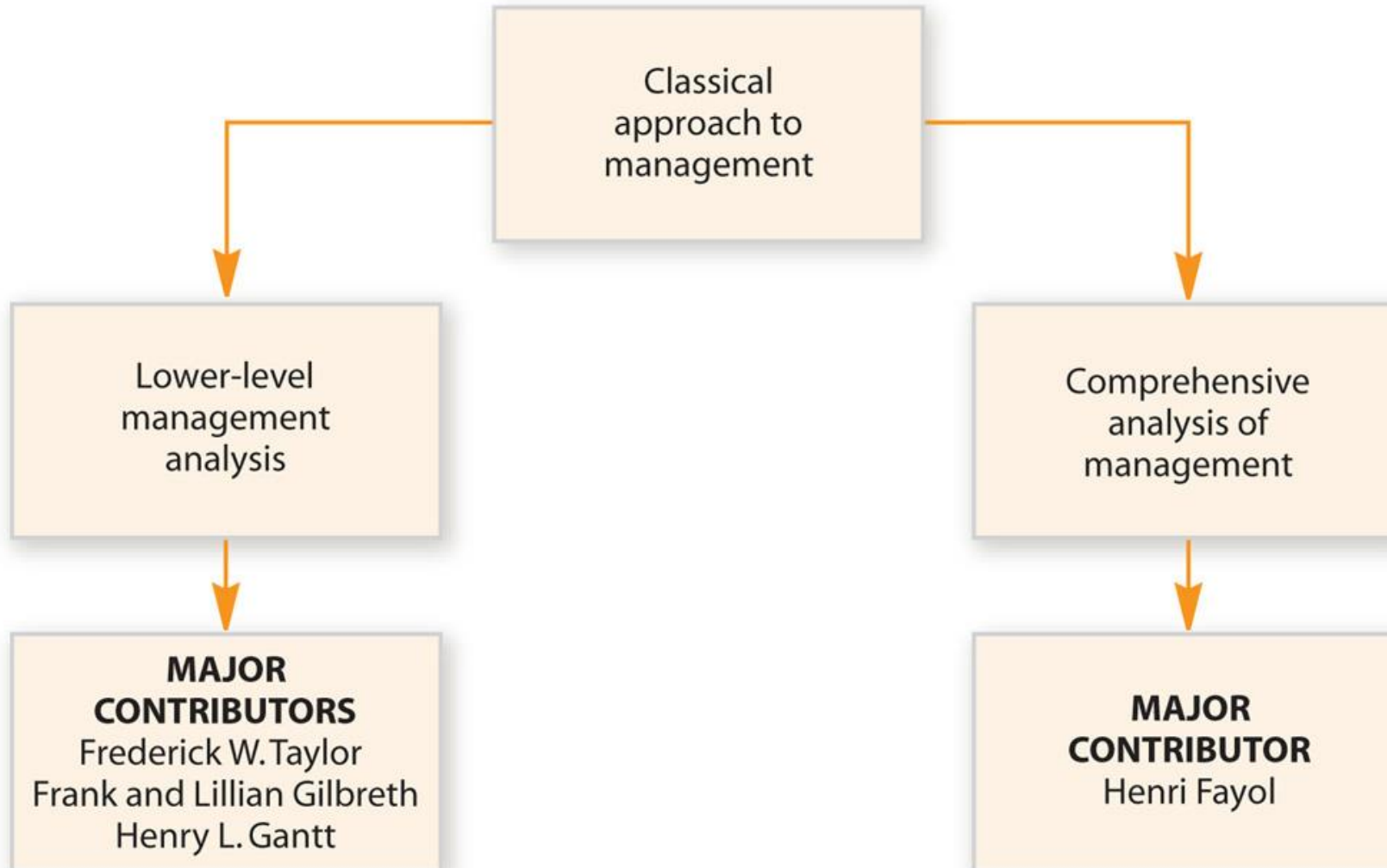
1. Classical Approach.
2. Behavioral Approach.
3. Management Science Approach.
4. Contingency Approach.
5. System Approach.
6. Learning Organization Approach.

Classical Approach

Is a management approach that emphasizes organizational efficiency to increase organizational success.



Classical Approach



Classical Approach - Low Level

Frederick W. Taylor (1856–1915) @ Bethlehem Steel Co.

- Father of Scientific Management
- Primary Goal: to increase worker efficiency by scientifically designing jobs
- There is only one best way to perform any specific task
- Price-Rate Rewarding System

Classical Approach - Low Level

Frederick W. Taylor (1856–1915) @ Bethlehem Steel Co.

Shoveling Science

1. Will a first-class worker do more work per day with a shovelful of 5, 10, 15, 20, 30, or 40 pounds?
2. What kinds of shovels work best with which materials?
3. How quickly can a shovel be pushed into a pile of materials and pulled out properly loaded?
4. How much time is required to swing a shovel backward and throw the load a given horizontal distance at a given height?

Classical Approach - Low Level

Frederick W. Taylor (1856–1915) @ Bethlehem Steel Co.

3 Years after applying science of shovelling:

- Number of shovelers needed no to do the same work

600 → 140 (%77 less)

- Average number of tones shoveled per worker per day

16 → 59 (%269 higher)

- Average earnings per worker per day

\$1.15 → \$1.88 (%63 higher)

- Average handling cost per a long ton

\$0.072 → 0.033 (%54 less)

Classical Approach - Low Level

Frank Gilbreth (1868–1924) and Lillian Gilbreth (1878–1972)

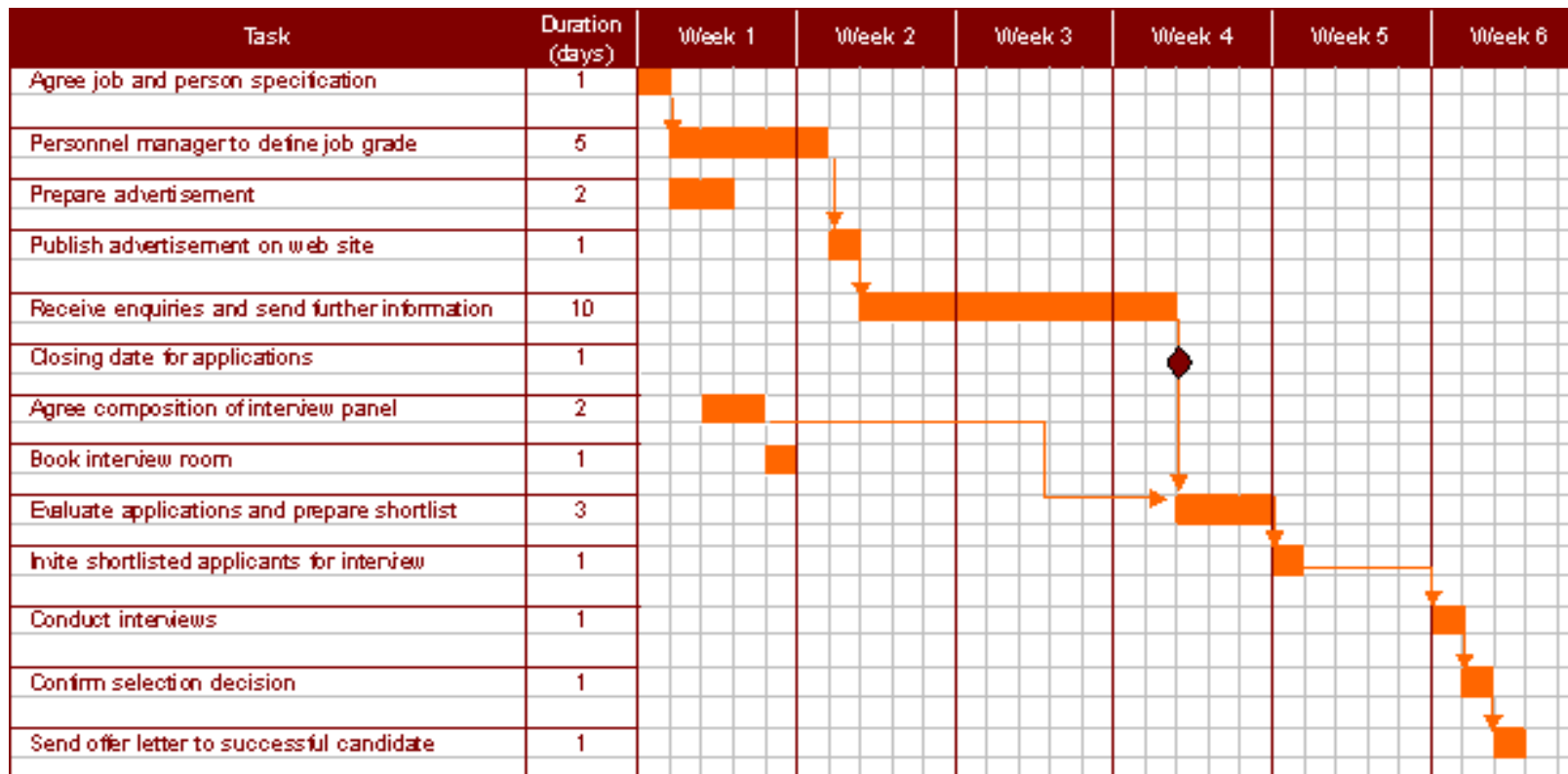
- Motion Study
- Unnecessary Movements to be eliminated.
- Lillian Gilbreth applied same concept in role of the housemaker and to the handicapped.



Classical Approach - Low Level

Henry L. Gantt (1861–1919)

- Rewarding Innovation (Bonus System)
- Scheduling Innovation



Classical Approach - Comprehensive

Henri Fayol (1841–1925)

- He wrote a book “*General and Industrial Management*”

- General Principles of Management:

- | | |
|--|---------------------------------------|
| 1. Division of work. | 8. Centralization. |
| 2. Authority. | 9. Scalar Chain. |
| 3. Discipline. | 10. Order. |
| 4. Unity of Command. | 11. Equity. |
| 5. Unity of Direction. | 12. Stability of tenure of personnel. |
| 6. Subordination of individual interests to the general interests. | 13. Initiative. |
| 7. Remuneration. | 14. Esprit de Corps. |

Classical Approach - Limitations

- Does not adequately emphasize human variables
- Bonuses doesn't seems to influence people nowadays as in the last century.
- Critical interpersonal areas were shortchanged such as:
 1. Conflict.
 2. Communication.
 3. Leadership.
 4. Motivation.

Mental Break

Behavioral Approach

- Emphasizes increasing production through an understanding of people
- Hawthorne Studies (1924-1932) @ Western Electric Company:
 1. The Relay Assembly Test Room Experiments
 2. The Bank Wiring Observation Room Experiment

Behavioral Approach

The Relay Assembly Test Room Experiments

- **Assumption:** Studying productivity long enough under different working conditions, would discover the working conditions that maximize production
- **Purpose:** to determine the relationship between intensity of lighting and worker efficiency, as measured by worker output
- **Experiment:**

Group 1



Group 2



Behavioral Approach

The Relay Assembly Test Room Experiments

- **Result:** No matter what conditions employees were exposed to, production increased.
- **Why?:** interviews had led to the followings:
 1. The subjects found working in the test room enjoyable.
 2. The new supervisory relationship during the experiment allowed the subjects to work freely, without fear.
 3. The subjects realized that they were taking part in an important and interesting study.
 4. The subjects seemed to become friendly as a group.

Behavioral Approach

The Relay Assembly Test Room Experiments

Conclusion

human factors within organizations could significantly influence
production

Behavioral Approach

The Bank Wiring Observation Room Experiment

- **Assumption:** members of a work group would pressure one another to work harder so that each group member would receive more pay
- **Purpose:** to analyse social relationships in a work group
- **Experiment:** The group piecework incentive system dictated that the harder a group worked as a whole, the more pay each member of that group would receive

Behavioral Approach

The Bank Wiring Observation Room Experiment

- **Result:** The work group pressured the faster workers to slow down their work rate
- **Why:** the men were more interested in preserving work group solidarity

Conclusion

Social groups in organizations could effectively exert pressure to influence individuals to disregard monetary incentives

Behavioral Approach

- Recognizing the Human Variable:
 - Human variable in the organization needed much more analysis.
 - Managers began to realize that they needed to understand this influence.
- The Human Relations Movement:
 - To enhance organizational success by building appropriate relationships with people.
 - To bring the effectiveness to the human relations inside the organization.

Mental Break

Management Science Approach

Beginning

- Operations Research in World War II
(Scientific Method + Mathematical Techniques).
- OR → MS (by UCLA-RAND group).
- The scientific method:
 1. Observe behavior
 2. Construct model
 3. Deduce new behaviors
 4. Test actual system

Management Science Approach

History

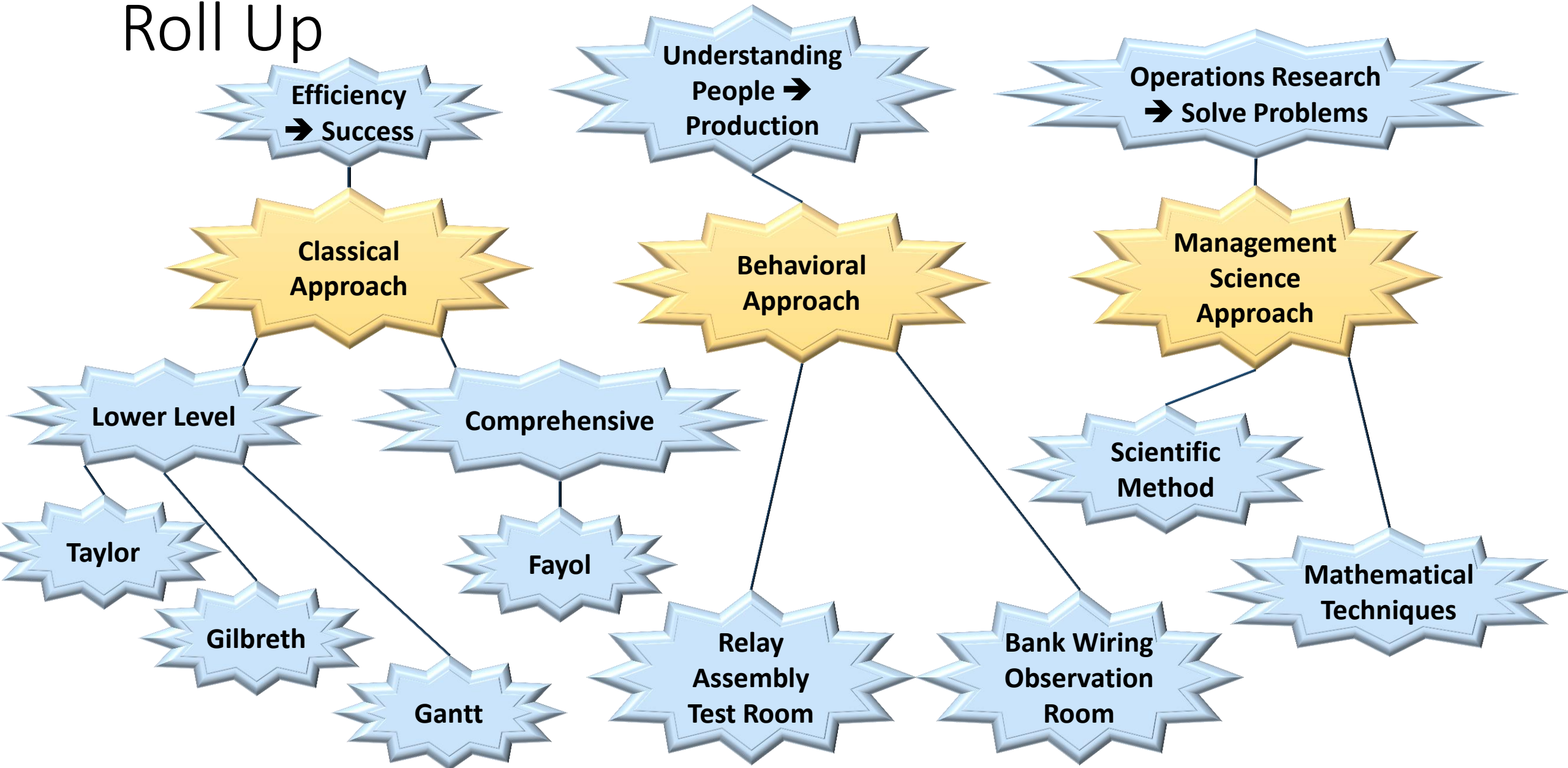
- By 1955: Effective in solving industrial problems.
- By 1965: Wide spread (Many companies + Many problems).
- By 1980: Extensively used in large & complex organizations (not small).
- 21st Century: Power of technology.

Management Science Approach

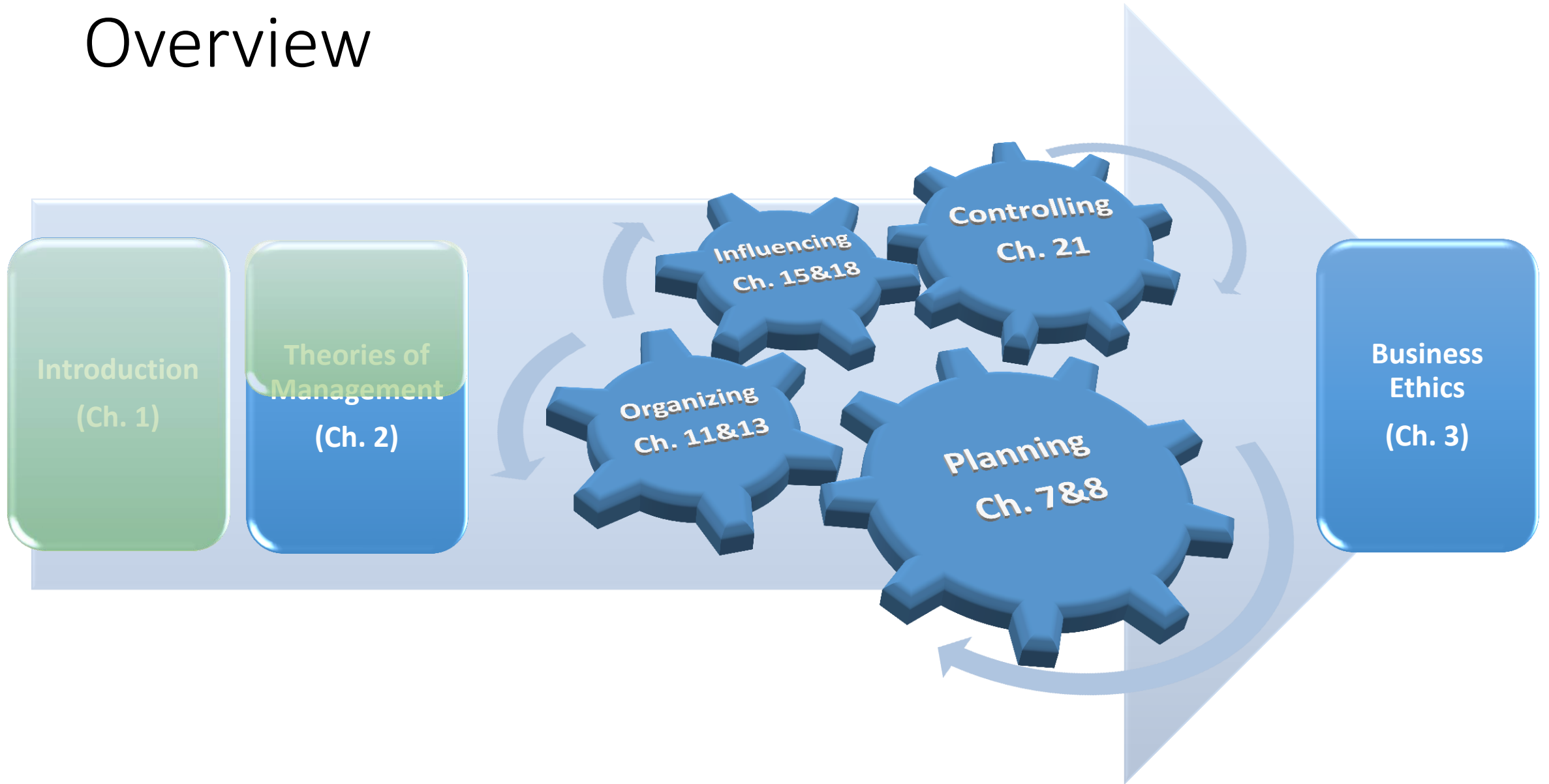
Characteristics of Management Science Applications

- Complicated Problems.
- Economic Implications.
- Mathematical Model.
- Computers & IT Usage.

Roll Up



Overview



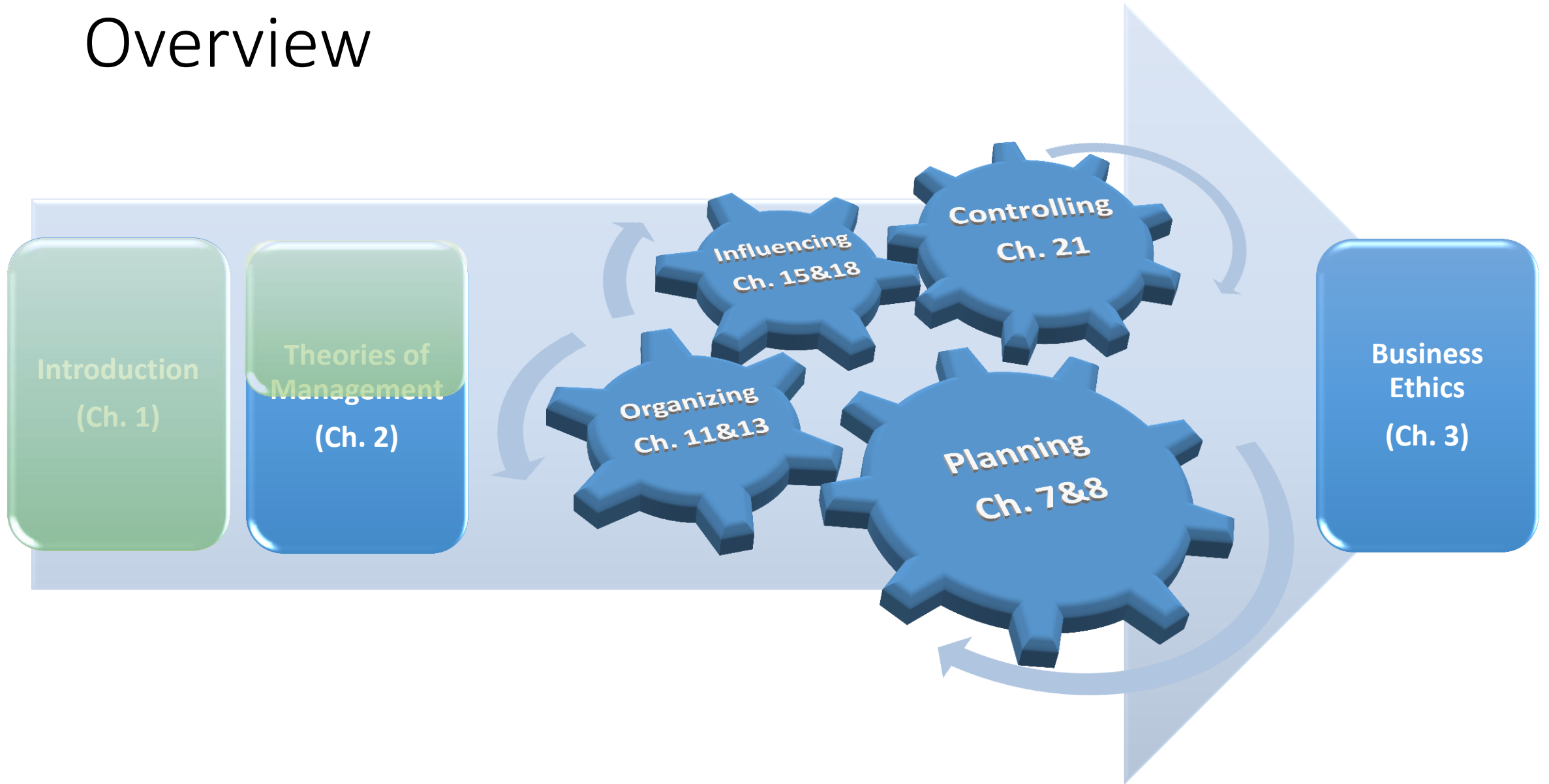
Chapter 2

History and Current Thinking

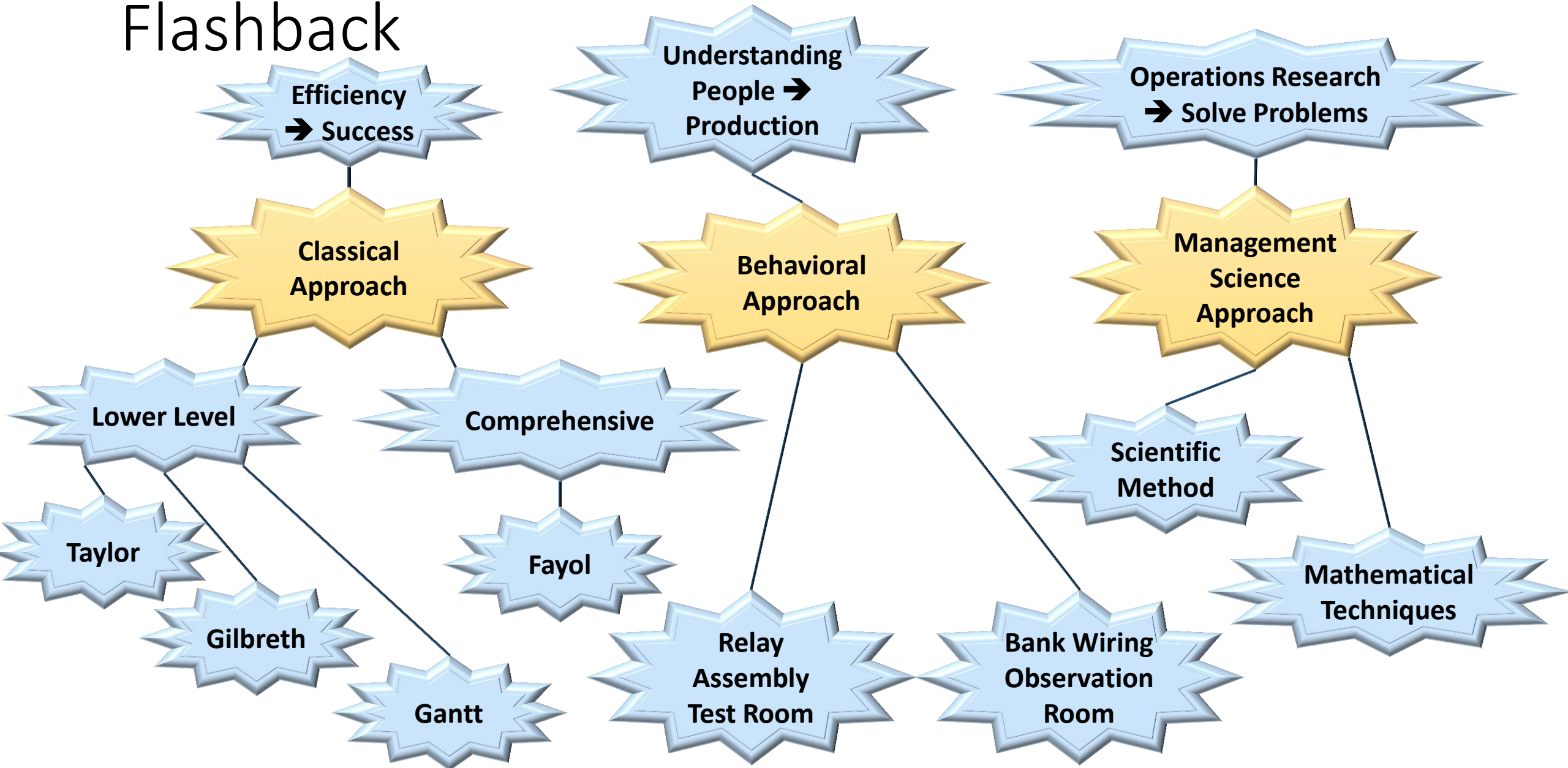
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Overview



Flashback



Objectives

1. *An understanding of the contingency approach to management*
2. *An understanding of the system types and system approach to management*
3. *Knowledge about the learning organization approach to management*
4. *An understanding of how triangular management and the contingency approach to management are related*

Management Approaches

1. Classical Approach.
2. Behavioral Approach.
3. Management Science Approach.
4. Contingency Approach.
5. System Approach.
6. Learning Organization Approach.

Contingency Approach

What managers do in practice depends on, or is contingent upon, a given set of circumstances

“If ... Then” Relationship

If a manager has a group of inexperienced subordinates, then the contingency approach would recommend that he or she lead in a different fashion than if the subordinates were experienced

Contingency Approach

It is an approach that attempts to outline the conditions or situations in which various management methods have the best chance of success

There is probably **no one** best way to solve **all** management problems in all organizations

There is probably **one** best way to solve **any given** management problem in any one organization

Contingency Approach

Challenges

1. Perceiving organizational situations as they actually exist
2. Choosing the management tactics best suited to those situations
3. Competently implementing those tactics

Mental Break

System Approach

General System Theory

- Founder: Ludwig von Bertalanffy (Physics & Biology Scientist)
- To fully understand the operation of an entity, the entity must be viewed as a system
- **System:** is a number of interdependent parts functioning as a whole for some purpose
- **Example:** Human Body {Ears, Eyes, Brain}
- System theory integrates the knowledge of various specialized fields so that the system as a whole can be better understood.

System Approach

System Types

- **Closed:** No influence and no Interaction with its environment is needed to achieve the purpose of the whole system.
- **Open:** Continually interact with its environment to achieve the purpose of the whole system.

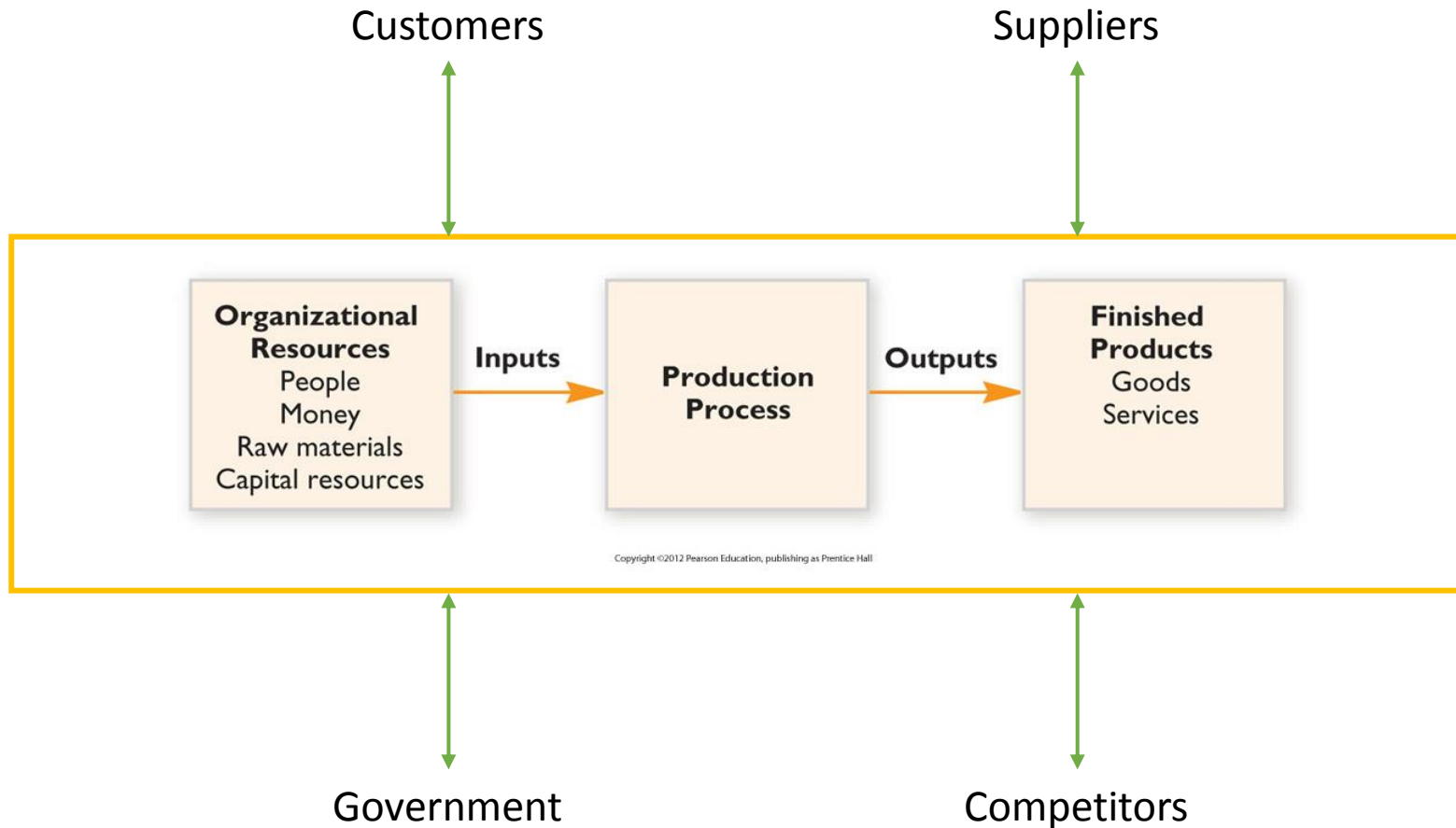
System Approach

System Wholeness

1. The whole should be the main focus of analysis
2. Integration is the key variable in wholeness analysis
3. Modifications weighed in relation to effects on every other part
4. Each part has some role to perform
5. Part and its function determined by its position in the whole
6. All analysis starts with the existence of the whole

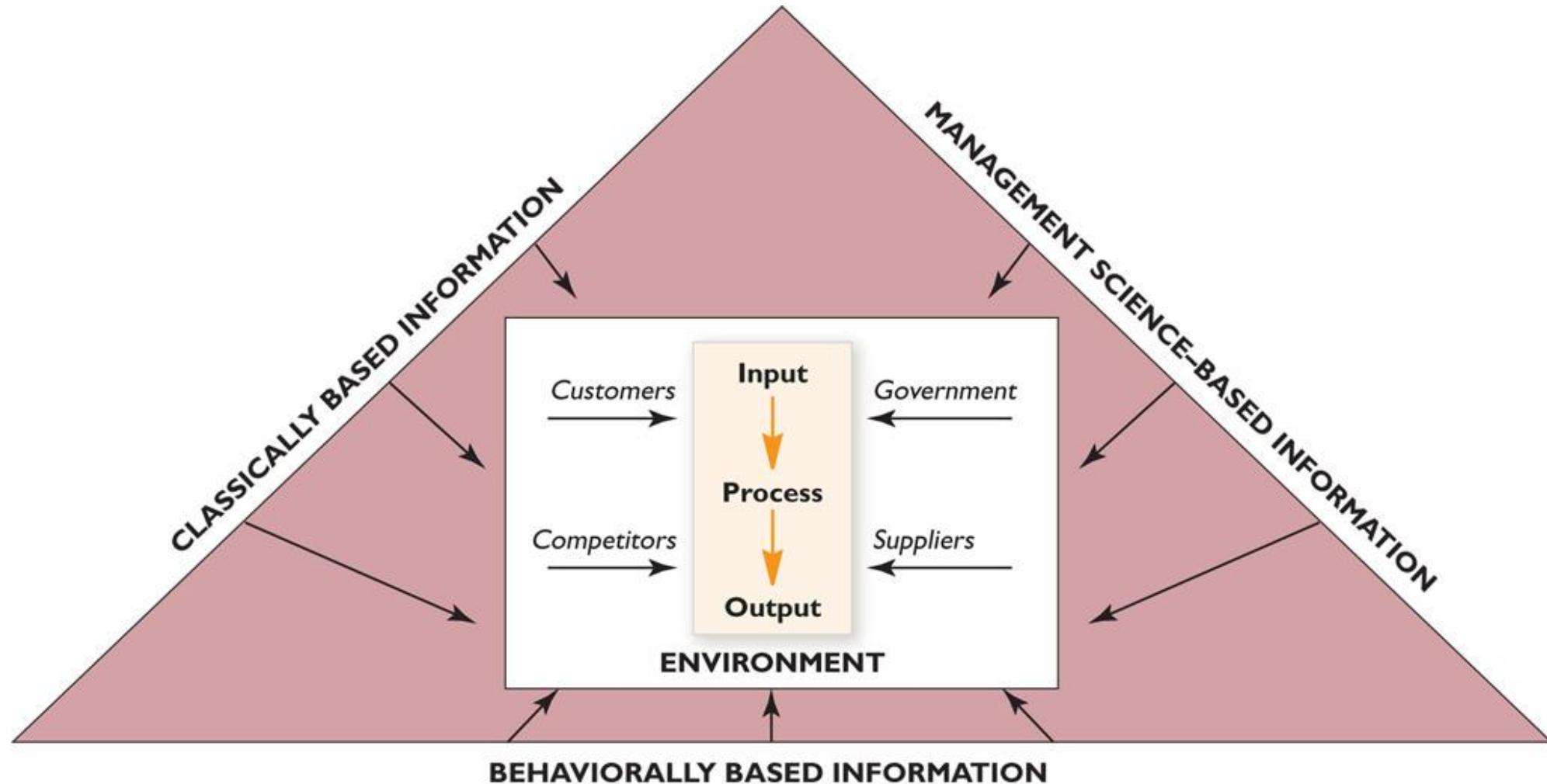
System Approach

Information for Management System Analysis



System Approach

Information for Management System Analysis - Triangular Management Model



Mental Break

Learning Organization: A New Approach?

Motivation

- Competitive decline of Western firms,
- Accelerating pace of technological change,
- Sophistication of customers,
- Increasing emphasis on globalization.

Learning Organization: A New Approach?

Motivation

- A **learning organization** is an organization that does well in creating, acquiring, and transferring knowledge, and in modifying behavior to reflect new knowledge

Learning Organization: A New Approach?

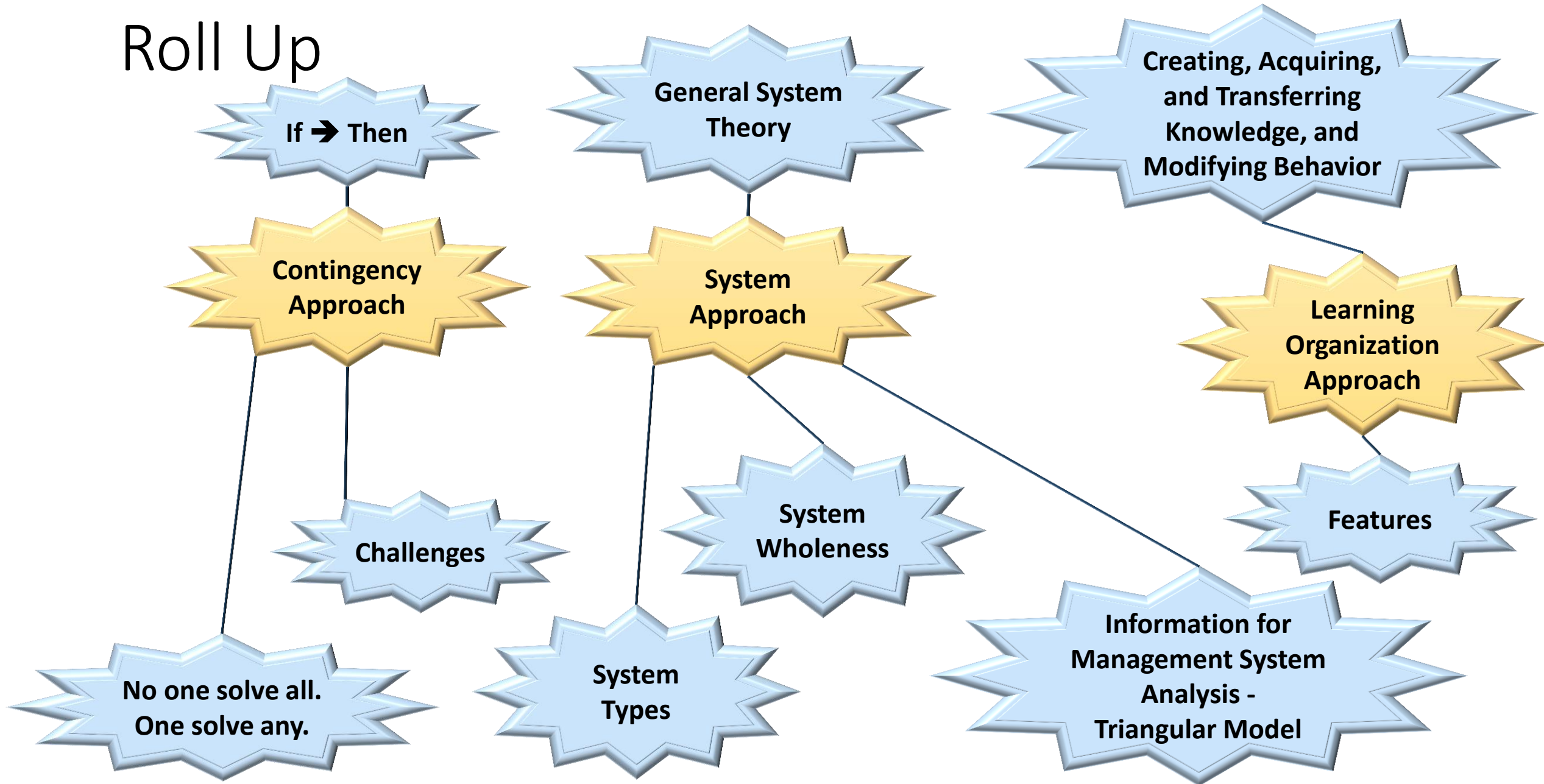
Learning Organization Features

1. Systems Thinking
2. Shared Vision
3. Challenging of Mental Models
4. Team Learning
5. Personal Mastery

Management Approaches

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Roll Up



Overview

